EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.
DATE:24/02/2026	

AIM:

To demonstrate session tracking using HttpSession and implement a simple login system where the user's session is tracked.

ALGORITHM:

1. **Input:**
    - The user fills out a web form with their name and email.
    - The form will use both GET and POST methods for data submission.

2. **Steps for Creating and Handling the HTTP GET and POST Requests:**

1. **Create an HTML Form:**
    - Design an HTML form with two input fields: one for the name and one for the email.
    - Create two buttons to allow the user to submit the form using GET or POST methods.
    - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).

2. **Create the Servlet to Handle GET and POST Requests:**
    - Write a Java Servlet class that will handle both GET and POST requests.
    - The servlet will have two methods:
    - doGet() to handle the GET requests.
    - doPost() to handle the POST requests.
    - In both methods, retrieve the form data using request.getParameter().
    - In the doGet() method, process and display the data in the URL.

- In the doPost() method, process and display the data in the HTTP request body.

3. Handle GET Requests in the Servlet:

- In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
- Output the data as a response, including the name and email in the URL (visible to the user).

4. Handle POST Requests in the Servlet:

- In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
- Output the data as a response, displaying the name and email without exposing them in the URL (more secure).

5. Return the Output:

- The Servlet should return an HTML page displaying the name and email entered by the user.
- For the GET method, the output should be visible in the URL.
- For the POST method, the output should be visible in the response body, not the URL.

6. Deploy and Test the Servlet:

- Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
- Test the form using GET and POST methods to see the difference in data submission.

7. Observe and Compare GET and POST Results:

- Test the GET method: The form data is sent in the URL.
- Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

PROGRAM:

```
<%@ page import="javax.servlet.http.*,java.io.*" %>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Session Tracking System</title>

<style>
:root {
   --sidebar-color: #2c3e50;
   --accent: #3498db;
   --success: #27ae60;
}

body {
   font-family: 'Segoe UI';
   margin: 0;
   display: flex;
   height: 100vh;
   background-color: #eef2f3;
}

nav {
   width: 240px;
   background-color: var(--sidebar-color);
   color: white;
   padding-top: 20px;
}

.nav-link {
   display: block;
   padding: 15px 25px;
   color: #bdc3c7;
}
```

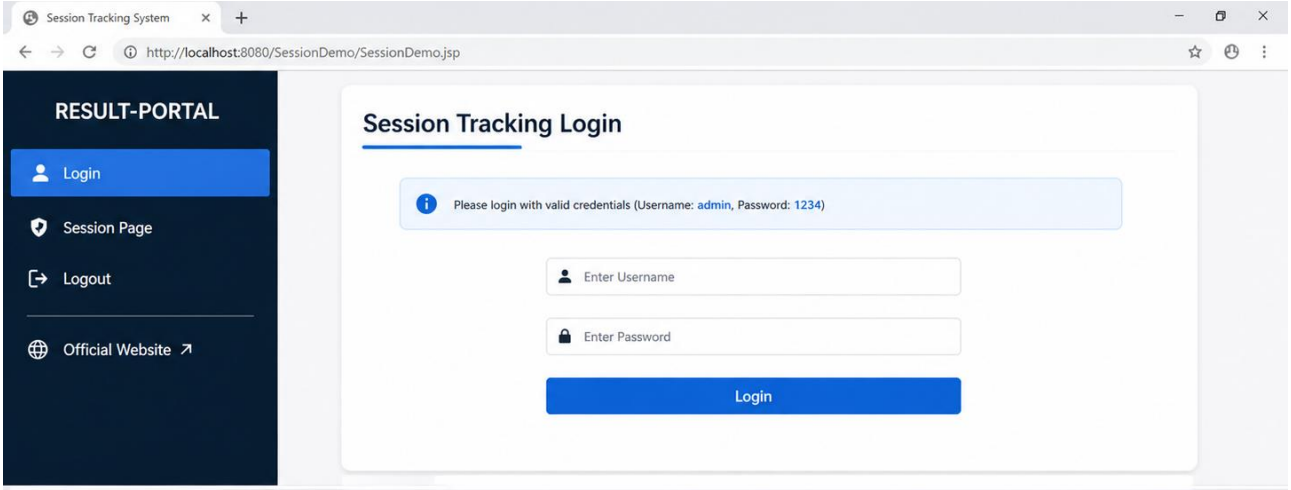
```
.nav-link:hover {
    background-color: #34495e;
    color: white;
}

main {
    flex: 1;
    padding: 30px;
}

.container {
    background: white;
    max-width: 600px;
    margin: auto;
    padding: 30px;
    border-radius: 12px;
}

input {
    width: 100%;
    padding: 10px;
    margin: 8px 0;
}

button {
    padding: 10px;
    width: 100%;
    background: var(--accent);
    color: white;
    border: none;
}

.success {
    color: var(--success);
    font-weight: bold;
}
.error {
```

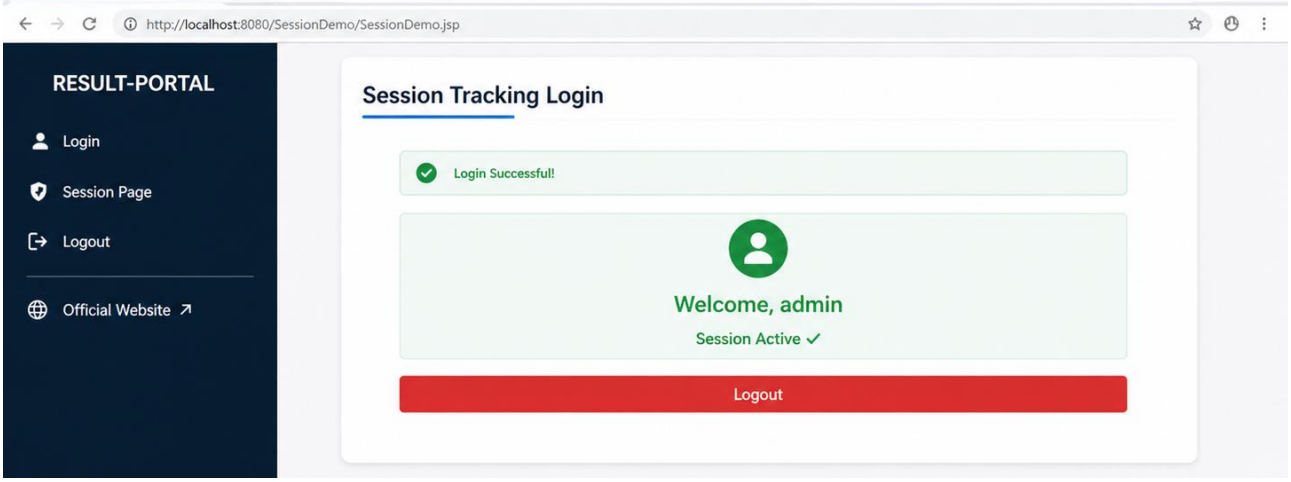
```
    color: red;
    font-weight: bold;
}
</style>
</head>

<body>

<nav>
    <h2 style="text-align:center;">RESULT-PORTAL</h2>
    <div class="nav-link">Login</div>
    <div class="nav-link">Session Page</div>
    <div class="nav-link">Logout</div>
</nav>

<main>
<div class="container">

<h1>Session Tracking Login</h1>

<%
    HttpSession sessionObj = request.getSession();
    String action = request.getParameter("action");

    if("logout".equals(action)) {
        sessionObj.invalidate();
%>
        <p class="error">Logged out successfully!</p>
<%
    }

    String username = request.getParameter("username");
    String password = request.getParameter("password");

    if(username != null && password != null) {
        if(username.equals("admin") && password.equals("1234")) {
            sessionObj.setAttribute("user", username);
```

```
%>
        <p class="success">Login Successful!</p>
<%
      } else {
%>
        <p class="error">Invalid Username or Password</p>
<%
      }
   }

   String user = (String) sessionObj.getAttribute("user");

   if(user == null) {
%>

<!-- LOGIN FORM -->
<form method="post">
   <input type="text" name="username" placeholder="Enter Username" required>
   <input type="password" name="password" placeholder="Enter Password" required>
   <button type="submit">Login</button>
</form>

<%
   } else {
%>

<!-- PROTECTED PAGE -->
<h2>Welcome, <%= user %></h2>
<p class="success">Session Active ✓</p>

<form method="post">
   <input type="hidden" name="action" value="logout">
   <button type="submit">Logout</button>
</form>

<%
   }
```

```
%>

</div>
</main>

</body>
</html>
```

Output:





RESULT:

A Servlet is created to demonstrate session tracking using HttpSession and implemented a simple login system where the user's session is tracked.